Healthcare System

Using Agile Methodology

1. Abstract

A Healthcare System is a digital platform designed to streamline medical operations, patient management, and appointment scheduling. In the modern medical landscape, hospitals and clinics require efficient, error-free, and accessible systems to ensure patient safety and operational effectiveness. This project focuses on developing a Healthcare System using the Agile Software Development Methodology. Unlike traditional waterfall models, Agile allows for iterative development, enabling the team to release features like patient registration, doctor portals, and e-prescriptions in small, manageable sprints.
The traditional manual handling of medical records is prone to errors, data loss, and slow retrieval times. Agile methodology addresses these challenges by fostering continuous collaboration between developers and medical staff (stakeholders). Feedback is gathered after every sprint to refine features, ensuring the software meets the real-world needs of doctors and patients.

The Healthcare System enables patients to book appointments online, doctors to view medical history and prescribe medication, and administrators to manage hospital resources. Agile ceremonies such as sprint planning and daily stand-ups ensure that the development remains on track and adaptable to changes. The result is a secure, user-friendly, and scalable application that enhances the quality of care, reduces administrative burdens, and ensures data integrity.

2. Introduction

2.1 Introduction

The Healthcare System automates hospital workflows, from patient admission to discharge. Agile methodology is employed to ensure the system is flexible, reliable, and user-centric.

2.2 Problem Identification

Manual record-keeping leads to lost files, slow patient processing, and lack of immediate access to critical medical history during emergencies.

2.3 Need of the Project

A centralized, digital system is needed to securely store patient data, manage doctor schedules, and provide 24/7 access to medical records.

2.4 Project Scheduling

Phase	Duration
Planning	2 Days
Development	8 Days
Testing	3 Days
Review	2 Days
Documentation	1 Day

2.5 Objectives

- Digitize patient health records (EHR).
- Streamline appointment booking.
- Use Agile methodology for adaptive development.
- Ensure data privacy and security.
- Reduce patient waiting times.
- Automate billing and prescription generation.

3. Software Requirement Specification (SRS)

3.1 Purpose

To develop a secure and efficient Healthcare Management System using Agile methodology.

3.2 Scope

Useful for hospitals, clinics, doctors, nursing staff, and patients.

3.3 Hardware / Software Requirement

Hardware:

- 4GB RAM
- Intel i3 processor or above
- 500GB HDD / SSD

Software:

- Windows 10 / 11
- Java / Web Technologies (HTML, CSS, JS)
- MySQL Database
- VS Code / Eclipse

3.4 Tools

- HTML, CSS, JavaScript (Frontend)
- Java / Spring Boot (Backend)
- MySQL (Database)
- Git (Version Control)
- Jira / Trello (Agile Management)

3.5 Software Process Model

- **Agile Model**
- Sprint Planning
- Daily Stand-up Meetings
- Sprint Review
- Sprint Retrospective

4. System Design

4.1 Data Dictionary

Field	**Type**	**Description**
PatientID	int	Unique Patient ID
DoctorID	int	Unique Doctor ID
AppointmentDate	Date	Date of visit
Diagnosis	String	Medical condition
Prescription	String	Prescribed medicines
BillAmount	double	Total cost of service

4.2 ER Diagram

Patient → Appointment → Doctor → Medical Record

4.3 DFD

Patient Details ⟶ Healthcare System ⟶ Database ⟶ Appointment Confirmation

4.4 Diagrams

- Use Case Diagram (Actors: Patient, Doctor, Admin)
- Activity Diagram (Appointment Process)
- Flowchart

5. Implementation

5.1 Program Code

Modules developed in sprints:

- **Sprint 1:** Patient Registration & Login
- **Sprint 2:** Doctor Availability & Scheduling
- **Sprint 3:** Appointment Booking System
- **Sprint 4:** Electronic Health Records (EHR) & Prescriptions
- **Sprint 5:** Billing & Invoicing

5.2 Output Screens

- Patient Dashboard
- Doctor Profile View
- Appointment Booking Form
- Digital Prescription View
- Billing Receipt

6. Testing

6.1 Test Data

Input	**Expected Output**
Valid Patient ID	Access Medical History
Double Booking	Error: Slot Unavailable
Empty Diagnosis	Validation Warning

6.2 Test Result

Unit testing and user acceptance testing (UAT) were conducted after each sprint to ensure system reliability.

7. User Manual

7.1 How to Use

1. **Register/Login** as a Patient or Doctor.
2. Patients: Select Doctor and click **"Book Appointment."**
3. Doctors: View **"My Appointments"** and update patient records.
4. Admin: Generate **"Daily Reports"** for hospital management.
5. Download **Prescription** or **Bill** after consultation.

7.2 Screen Layout

Clean, accessible interface with distinct portals for Patients, Doctors, and Admin.

8. Project Applications & Limitations

8.1 Applications

- Hospital Management
- Private Clinics
- Telemedicine Services
- Pharmacy Integration

8.2 Limitations

- Dependency on internet connectivity.
- Data security regulations (e.g., HIPAA) compliance is complex.
- Elderly patients may require assistance using the digital interface.

9. Conclusion & Future Enhancement

Conclusion:

The Healthcare System developed using Agile methodology provides a vital solution for modernizing medical facilities. The iterative Agile approach ensured that critical features were delivered quickly and that the system remained adaptable to the specific needs of medical staff.

Future Enhancements:

- AI-based Symptom Checker.
- IoT integration for remote patient monitoring (wearables).
- Video Consultation (Telemedicine).
- Blockchain for secure medical records.

10. Bibliography & References

- Agile Manifesto
- Healthcare Information and Management Systems Society (HIMSS)
- Java Programming Documentation
- GeeksForGeeks
- IEEE Papers on E-Health